

thresholds, and diagnostic features can be configured through software, and adapting the system to different models often requires only a reflash. This reduces the need for multiple hardware variants and improves inventory control.

At the same time, the input side benefits from reduced harness complexity in both connection count and current levels. Higher-voltage operation leads to efficiency gains of 7 to 12% due to reduced current and lower conduction losses. The LCM concept supports the need for a complete software-based lighting system design.

Software-defined lighting platforms

From a hardware perspective, one of the biggest advantages of this approach is its flexibility. Adapting the system for different vehicle models generally requires only a software reflash, enabling efficient inventory control. A single platform can be reused with programmable current levels and configurable operating parameters, eliminating the need to design, source, or manage multiple hardware variants.

Fewer connectors and synchronous circuit topologies also shrink PCB space requirements. Higher efficiency reduces thermal dissipation, removing the need for large heat-sinks and allowing a more compact design. Component support for such architectures is already available: Texas Instruments' LM708x0-Q1 family supports up to 80V input with 4A to 8A current output, suitable for four-wheeler lighting, while Monolithic Power Systems offers compact devices rated up to 90V with 0.9A to 3.5A output current suitable for two-wheelers. TI's LMR38025 works well for first-stage conversion.

For the second stage, drivers like LP8865 can be used for cost-effective 65V, 2A operation, or TPS92543 for advanced adaptive control with multiple LEDs in series, bypass dimming, or dynamic

effects. Multi-channel drivers further simplify the management of multiple LED strings. For lower-current loads such as turn indicators, linear drivers can replace switching regulators, reducing complexity.

The hardware components work with microcontrollers supporting 8-bit, 16-bit, and 32-bit architectures, which manage CAN and LIN networks, process inputs, and control LED drivers. The architectural design delivers a simple structure that combines advanced intelligent capabilities with efficient performance and scalability, enabling the management of various vehicle systems and lighting requirements through software-based control.

Tesla is setting the 48V bar

Even from its form factor, this architecture can be readily associated with Tesla, Inc.. The Tesla Cybertruck has effectively replaced the standard 12V battery with a 48V system. All lighting loads are powered directly from this setup. In a typical case, the front lighting system uses about 175W, which is around 3.35A at roughly 52V. Notably, this current level is similar to that of electric two-wheelers like the Bajaj Chetak, Ather 450X, or TVS iQube, which draw power from a 12V battery for their lighting systems. Just imagine the simplification in connector selection and the reduction in connector size due to lower current demand.

In summary, as illustrated through the lighting example, a wide range of IC solutions are already available across applications, whether in two-wheelers or four-wheelers, covering loads such as lighting, window winders, audio systems, and instrument clusters. Just as the industry once transitioned from 6V to 12V, the ecosystem is now well positioned for a shift from 12V to 48V, and this transition is most effective when adopted holistically rather than in part.

From a performance standpoint,

the benefits are substantial. Overall system efficiency can improve by 7.5-12%, while input current can be reduced to nearly one-third in worst-case scenarios and typically to one-fourth. In addition, since a significant portion of the load, nearly 30% from lighting alone, can be offloaded, DC-DC converter sizing can be reduced, leading to savings in both cost and space. The use of more integrated, synchronous components further reduces PCB footprint and eliminates the need for external MOSFETs.

By using a platform-based design, a single hardware solution can work across multiple vehicle models or OEMs with minimal tweaks. The result is a system that reduces costs at the ECU (electronic control unit), harness, and connector levels, while being more efficient, compact, scalable, and ready for the future of automotive design.

The 48V horizons

This migration enables the industry to address several critical objectives simultaneously, with miniaturisation emerging as a key driver. Across electronics, from mobile devices to advanced test equipment, form factors are steadily shrinking. Even display instruments like oscilloscopes, once bulky, are now increasingly compact, reflecting the broader direction of the ecosystem.

As this evolution continues, system design priorities must align accordingly, balancing miniaturisation with efficiency and cost optimisation. This ensures future architectures are not only more compact but also energy-efficient and economically viable. In this context, I am keen to collaborate with anyone interested in early-stage prototyping or exploring similar architectures. **EFY**

The article is based on the talk at the EFY Expo Pune 2026 titled Rethinking Automotive Lighting for the 48V Era, featuring a speech by Mahendra Patel, Electronic Design Consultant, AIGHTECH Services Pvt Ltd, India. It has been transcribed and curated by Saba Aafreen, Tech Journalist at EFY.